

Differential Location Analysis of Transcription Factors

-Capstone Project Proposal Draft-

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Introduction

It's widely known that gene expression programs depend on recognition of specific promoter sequences by transcriptional regulatory proteins so called transcription factor (TF). Knowledge of the sites bound by all the transcriptional regulators encoded in a genome can provide the information necessary to nucleate models for transcriptional regulatory networks especially for complex eukaryote systems which employs complicated cooperation among many different TFs.

Background

With the advent of Chromatin Immunoprecipitation on microarray, or so called ChIP-on-Chip, along with the availability of complete genome sequences, genome-wide location analysis has become one of the hot areas in genomics researches. There has been a recent research to identify all cis-regulatory regions bound by 203 yeast transcription factor with different culturing condition(1). They have proposed different promoter architectures regarding the binding sites of each transcription factor and showed that many transcription factors have different binding affinity to same cis-regulatory region depending on the experimental condition. In the respect of gene regulation, this makes sense since required sets of genes would be different in different condition. However, they only used qualitative measures using only p-values as threshold for determination of TF bound or not. Here we want to exploit these hidden mechanisms of transcriptional regulation and cooperation among different TFs.

Goal

By using ChIP-on-chip data as well as available microarray gene expression, protein-protein interaction, motif analysis, we are expecting to reveal underlying correlation between different transcription factors in a quantitative measure.

Methods

1. Identification of differentially bound promoter regions on different culturing conditions.
 - A. Simple correlation
$$\text{Corr} = (A - B) / (A + B) \quad (A, B : \text{binding affinity of TF in different condition})$$
$$\text{Corr} \geq 0.5 \text{ or } \text{Corr} \leq -0.5 \text{ will be selected for further analysis.}$$
2. Comparison with different sources
 - A. Microarray gene expression data to identify any TF showing altered expression level in different condition.
 - B. ChIP-on-Chip binding results to identify the most-influencing other TF(s) which have cis-regulatory binding region at the same promoter.
 - C. Comparison with Protein-Protein interaction to identify TF(s) which may interact
 - D. Motif discovery analysis to identify novel motif.

Expected outcomes

From the proposed project, we are expecting to reveal the quantitative relationship between eukaryotic transcription factors.

References

1. Harbison, C.T. et al, Transcriptional Regulatory Code of a Eukaryotic Genome. Nature 431: 99-104 (2004).